

KAIHUAI DENG

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EDUCATION

Nanjing University of Information Science and Technology (NUIST)

Nanjing, China

B.S. in Atmospheric Science (Hydrometeorology)

Sep 2023 - Jul 2027 (Expected)

NUIST ranked #1 in China and #8 globally for Atmospheric Science (2025 ShanghaiRanking GRAS).

- Weighted average: 85.53/100 | GPA: 3.57/4.0 | Class rank: 6/51 (**top 12%**)
- Undergraduate researcher since 2023, Key Laboratory of Hydrometeorological Disaster Mechanism and Warning of Ministry of Water Resources.
- Relevant coursework: Probability & Statistics, Remote Sensing for Hydrometeorology, Hydrometeorological Modeling, Numerical Weather Prediction, Python Programming.

RESEARCH INTERESTS

Remote Sensing · Hydroclimate Extremes · Land–Atmosphere Interactions · Climate Risk Analysis

MANUSCRIPTS & PRESENTATIONS

[1] **Deng, K.**, Huang, Z.* *Area Expansion Dominates Rising Global Three-Dimensional Terrestrial Water Storage Drought Severity*. Manuscript in preparation; presented in English at the 2026 National Outstanding Undergraduate Forum, Department of Earth System Science, Tsinghua University (**70 participants nationwide**).

[2] **Deng, K.**, Huang, Z.*, Li, H., Wu, J. *Global Unequal Exposure to Intensifying Hydrological Droughts*. Manuscript under review at *Journal of Hydrology*; third revision submitted.

[3] Wang, J.*, **Deng, K.** (2024). *A Model that Explains the Contrasting SST Trends in the Southern Pacific Ocean*. *STECEP*. <https://doi.org/10.61173/xnag2k18>

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RESEARCH EXPERIENCE

Mitacs Globalink Research Internship

Charlottetown, PE, Canada

Research intern | Advisor: Prof. Xander Wang (University of Prince Edward Island)

Jul 2026 - Present

- Awarded the fully funded 2026 Chinese Scholarship Council (CSC)-Canada Mitacs Globalink Research Internship (**≤260 awardees nationwide**).
- Analyzing land-atmosphere coupling and GRACE-based groundwater trends for Prince Edward Island; comparing satellite terrestrial water storage estimates with local well observations to assess freshwater sustainability.

Global 3D Hydrological Drought Tracking & Mechanism Attribution

Sep 2025 - Present

Independent research | Advisor: Prof. Zhongwei Huang (NUIST)

- Developed a three-dimensional event-tracking framework for 23 years of GRACE/GRACE-FO terrestrial water storage observations, identifying 672 drought events and reconstructing their splitting–merging lineages.
- Designed an evaporative-contribution diagnostic to separate precipitation-deficit and evapotranspiration-related influences and quantify how spatial expansion contributes to increasing event severity.
- Principal Investigator of a Provincial Undergraduate Training Program on Innovation and Entrepreneurship.

Global Unequal Exposure to Intensifying Hydrological Droughts

Jun 2025 - Aug 2026

Research assistant | Advisor: Prof. Zhongwei Huang (NUIST)

- Applied EEMD to GRACE/GRACE-FO CSR RL06 mascon observations and derived event-based drought metrics; integrated GPW population grids to estimate a global trend of 38.76 million person-exposures yr⁻¹.
- Decomposed the exposure increase into hydroclimatic conditions (73.2%), population growth (5.3%), and their interaction.
- Led study design, global data processing, visualization, manuscript drafting, and reviewer-response analysis under faculty supervision; third revision submitted to *Journal of Hydrology*.

Independent research | Advisor: Prof. Jun Yin (NUIST)

- Extended a nonequilibrium fluctuation theorem (Yin et al., 2024, *J. Climate*) from surface temperature to TCWV using ERA5; quantified PDF asymmetry A_{asym} across $\tau = 1\text{--}20$ yr smoothing timescales.
- Validated SST–TCWV Clausius–Clapeyron scaling ($R^2 = 0.95$, approximately 7%/K) and tested the transferability of the tail-rescaling framework; produced a manuscript draft; discussed with Prof. Lamberto Rondoni (Politecnico di Torino).

WRF-based Precipitation Sensitivity over Jing-Jin-Ji

Mar 2025 - Jul 2025

Independent research | Advisor: Dr. Mengya Wang (NUIST)

- Configured WRF/WPS with three nested domains on Linux HPC; conducted 16 experiments perturbing soil moisture ($\pm 10\%$ to $\pm 50\%$) and urban LULC to represent rapid-urbanization scenarios.
- Evaluated precipitation responses using spatial correlation, RMSE, and threat scores; identified the lowest-bias cumulus–microphysics configuration among the tested experiments.

Precipitation–Runoff Prediction in the Yiluo River Basin

Jun 2024 - Mar 2025

Research assistant | Advisor: Prof. Xing Yuan (Institute of Atmospheric Physics, Chinese Academy of Sciences)

- Applied EOF to monthly precipitation interannual increments, extracting three PCs explaining $>90\%$ of variance; screened hydroclimatic predictors using correlation analysis and stepwise regression.
- Built a BiLSTM + interannual-increment hybrid model, improving TCC by 16% and reducing RMSE by 20% relative to statistical downscaling; evaluated CSSPv2 monthly streamflow forecasts.

Contrasting SST Trends in the Southern Pacific Ocean

Mar 2024 - Sep 2024

Student researcher | Advisor: Prof. George Tselioudis (Columbia University / NASA GISS)

- Analyzed 29 years of multi-source satellite and reanalysis data (Copernicus SST, DUACS SSH, CERES, NCEP, NSIDC SIC) across the Southern Pacific ($80^\circ\text{W}\text{--}180^\circ$, $70^\circ\text{S}\text{--}30^\circ\text{S}$).
- Derived Ekman transport (U_E, V_E, W_E) from wind stress to attribute the SST dipole and diagnosed a sea-ice radiative positive-feedback cycle; published as co-author in *STECEP* (2024).

INTERNSHIP

Leizhou Meteorological Bureau, Guangdong

Jul 2025 - Aug 2025

Intern | Forecast Division

Zhanjiang, China

- Supported operational monitoring and warning for Typhoon Danas (2025) and Typhoon Wipha (2025), tracking intensification and landfall trajectories and comparing ECMWF, CMA, NCEP, and TRAMS forecasts.
- Participated in calibration and telemetry checks for ometeorological field instruments.

SELECTED AWARDS & HONORS

Provincial Undergraduate Training Program: Principal Investigator (provincial-level project) May 2026

Mitacs Globalink Research Internship Award: CSC–Mitacs Joint Program (≤ 260 awardees nationwide) Jan 2026

Second Class Academic Scholarship: NUIST (top 20%) Nov 2024 & Nov 2025

Merit Student: NUIST (top 15%) Dec 2024

ADDITIONAL INFORMATION

Programming: Python, Shell scripting, Fortran, Typst/LaTeX

Scientific Computing: Google Earth Engine, Linux/HPC, Git, WRF/WPS, MATLAB, ArcGIS

Machine Learning: TensorFlow, PyTorch, BiLSTM, EOF, EEMD

Languages: Mandarin (native), Cantonese (native), English (proficient), French (basic)

Academic Service: Creator and maintainer of othodeng.github.io, publishing study notes from 10+ courses